for Library

01430

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 3, Semester 2 (2025)

IT3060 – Human Computer Interaction

Duration: 2 Hours

November 2025

Instructions to Candidates:

- This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 5 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 01 (20 Marks)

Human-Computer Interaction (HCI) is a discipline concerned with how people use, experience, and interact with computer-based systems.

a) Name four (04) UX design principles/factors in Human-Computer Interaction (HCI). **(2 Marks)**

b) Define the concept of 'Context' in HCI. Briefly explain two (02) out of 5 types of contexts in HCI. **(4 Marks)**

c) Analyze the difference between user interface (UI) design and user experience (UX) design. Use a banking application as an example to support your analysis. **(4 Marks)**

d) State whether you Agree or Disagree with the statement below. Justify your answer in relation to the user experience (UX) domain. **(3 Marks)**

Design is not just what it looks like and feels like. Design is how it works.

- Steve Jobs

e) Identify two (02) possible accessibility issues /disabilities expecting as an UX designer. Recommend two (02) design suggestions for each identified accessibility issue. **(4 Marks)**

f) Explain what the 'Think Aloud" Evaluation method is and its significance. **(3 Marks)**

Question 02 (30 Marks)

"StudyMate" is an intelligent and human-centered platform designed to assist undergraduate students in managing their academic, emotional, and social well-being. It integrates academic planning, collaborative learning, and self-care tools within a single user-friendly environment. This app enables users to plan study sessions, set academic goals, and track their progress through interactive dashboards and AI-based analytics.

To address mental and emotional well-being, "StudyMate" incorporates mood tracking and guided reflection features that help students identify stress triggers and receive personalized coping strategies. This app also provides quick relaxation exercises and reminders to take healthy study breaks by promoting mental resilience during demanding academic periods. Recognizing the value of peer connection "StudyMate" encourages collaboration through virtual study groups, discussion boards, and real-time mentoring sessions. These community features allow students to share notes, clarify concepts, and build meaningful academic networks that enhance motivation and belonging.

Assume you have been assigned to the design team tasked with creating a mobile app for "StudyMate".

a) The Design Thinking Process is a structured framework that UX designers utilize to guide and organize their design approach. Apply design thinking model for the "StudyMate" scenario and briefly describe each phase. **(5 Marks)**

b) Identify two (02) user groups for the "StudyMate" app and provide a brief description of each group. **(4 Marks)**

c) Suggest two (02) suitable user research methods to collect user needs related to '**mental and emotional well-being**'. Justify your suggestions. **(5 Marks)**

d) Explain the concept of Participatory Design and describe two (02) ways end users and key stakeholders could actively contribute to the participatory design process of the "StudyMate" application. Justify how their involvement could enhance the app's usability and user satisfaction. **(6 Marks)**

e) Apply three (03) dimensions of interaction design to describe how each selected dimension can enhance the user experience in the "StudyMate" app's mood-tracking or study-planner features. Support your answer with suitable examples for each. **(6 Marks)**

f) Identify two (02) suitable interaction types that can be applied to the StudyMate app. Explain how each supports user interaction and improves usability. **(4 Marks)**

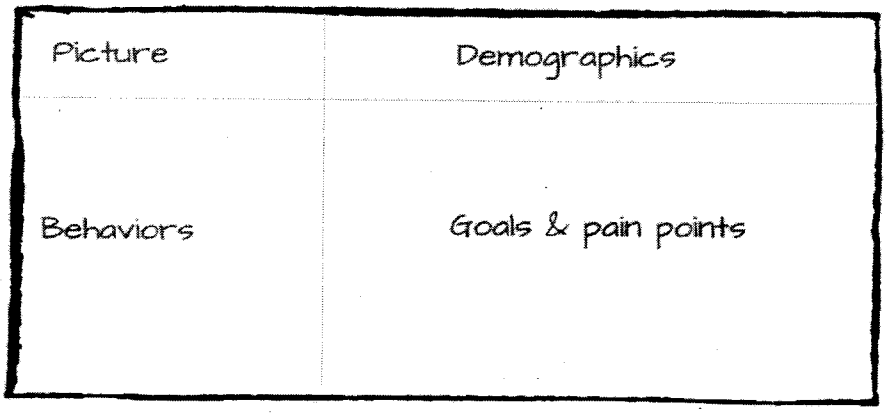
Question 03 (30 marks)

"**TalentTree**" provides professionals with a structured way to plan and track their skill development goals. Through the app, users can schedule learning sessions, enroll in short online courses, and access mentorship from industry experts. The platform tailors course recommendations based on career goals, available time slots, and learning styles. "TalentTree" helps busy professionals continuously upskill without disrupting their work-life balance.

Assume that you have been appointed as the UI/UX engineer to design an application for the scenario mentioned above.

a) Identify three (03) user groups for the "TalentTree" app and provide a brief description for each group. **(4.5 Marks)**

b) Select one user group from part (a) and create hypothetical personas for the selected group. Use the template below to complete the personas. **(2.5 Marks)**



c) During the development of the "TalentTree" app, the design team is planning to apply hybrid brainstorming to generate new ideas for improving the platform's core features. Explain how this technique supports creativity and idea generation in the design process. **(3 Marks)**

d) Explain two (02) ideation techniques you would apply for the given scenario except the technique mentioned in Part (b). **(4 marks)**

e) As an UI/UX engineer of the "TalentTree" team, you are responsible for creating prototypes.

i. Briefly explain two (02) major variations of prototypes. **(4 Marks)**

ii. The "TalentTree" team wants to test the basic user flow of 'finding a recommended course' and 'scheduling a learning session' functionalities during the initial design phase. From the two variations presented in part (e) (i), recommend which variation allows you to achieve the above task. Justify your recommendation by providing two (02) examples for the selected variation. **(5 Marks)**

iii. The project manager suggests that the "TalentTree" team need to directly move to coded HTML/CSS prototype to save time. State whether you agree or disagree with project manager's idea. Justify your answer by providing potential risks and drawbacks or benefits and advantages as applicable. **(3 Marks)**

iv. A user with color blindness (deuteranopia) wants to use "TalentTree's" progress-tracking charts, which currently use red for "behind schedule" and green for "on track." Propose a redesign of this chart to make it accessible, specifying two (02) changes. **(4 Marks)**

Question 04 (20 Marks)

a) Differentiate formative evaluation and summative evaluation in UX design. **(4 Marks)**

b) "Heuristic Evaluation is an effective method for assessing the usability of interface." Explain the meaning of Heuristic Evaluation. **(3 Marks)**

"**SmartLearn**" provides online lessons, quizzes, and progress tracking for students. However, students often become frustrated because they are unsure whether their quiz submissions have been saved. Additionally, the layout and navigation structure differ between pages, causing disorientation.

c) Analyze the above scenario and explain how Feedback and Consistency principles can be effectively applied to improve the quiz submission process and overall navigation experience. **(6 Marks)**

d) Explain how the Visibility principle can be implemented to ensure students can clearly identify and efficiently use important options such as "Submit" and "Next Lesson." **(4 Marks)**

e) Suggest a design modification that prevents students from accidentally leaving a quiz page without saving their responses. **(3 Marks)**

- - End of Question Paper - -